Self-supervised Through-Plane Resolution Enhancement for Lung Phantoms Using UNets

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***Abstract*—Computed Tomography (CT) imaging is an essential diagnostic tool in clinical practice, offering non-invasive visualization of internal body structures. However, limitations such as low resolution in the through-plane direction and slice overlap due to helical scanning trajectories continue to hinder the accuracy and utility of CT data. Recent advancements in deep learning–based super-resolution techniques have shown promise in addressing these challenges, yet many approaches are constrained by assumptions like fixed resolution ratios or non-overlapping slices. Self-supervised learning also enables generalization on the current dataset without the need for large amounts of labeled data, unlike supervised learning. In this work, we build upon the SR4ZCT method, which leverages off-axis training and accurate interpolation to support arbitrary resolution and overlap. We adapt this methodology using a U-Net architecture and evaluate its effectiveness on custom-designed lung phantoms.**

***Index Terms*—Computational Tomography, Resolution Enhancement, Deep Learning, UNets**

I. Introduction

For several decades, Computed Tomography (CT) scans are used as a non-invasive solution [1] to visualize areas in our bodies and allow medical practitioners to gain an understanding of what is the situation in a patient's body without having to just rely on symptoms or accident descriptions. This technique relies on X-rays that go through a patient's body in a direction and then measured on the other side. Different types of tissue in the body absorb the radiation in different amounts, thus learning from different reductions of the signal power measured. These measurements can be processed by digital computers to provide grayscale images of the targeted area for the medical doctors. However, there are still limiting factors present in modern systems like lower resolution in the plane perpendicular to the direction of the imaging plane [2] or overlap between slices due to using a helical trajectory. Such factors can sometimes make the scans insufficient for a proper diagnosis [2],[3],[4].

Many modern deep learning techniques have been applied with the objective of tackling this issue. Supervised techniques make use of low and high resolution volumes [5] [6] and self-supervised techniques are applied in the case that the high resolution volumes are not available, simulating LR training images from HR in-plane images so as to train a super-resolution model to on the in-plane dimension and then apply it to the through-plane dimension in order to enhance the total resolution [7] [8]. Several such algorithms perform well, but only when there is no overlap or a specific resolution ratio. Our work is based on an algorithm that fixes these issues, SR4ZCT [9], which builds upon the concept of off-axis training and is able to handle arbitrary resolution/overlap using accurate interpolation techniques to generate the low resolution data.

In this paper, we present an adaptation of the SR4ZCT method, replacing its original architecture with a U-Net model. In Section 2 (Method), we introduce the necessary mathematical notation, review the original SR4ZCT algorithm, and describe our modifications. Section 3 (Experiments) details the experimental setup, including the generation of synthetic lung phantoms, the metrics used for evaluation (PSNR, SSIM), and a description of our experiments. In Section 4 (Results), we present both visual and quantitative results of our method. Section 5 (Discussion) explores key findings, discusses the advantages and limitations of our approach, and outlines potential directions for future work. Finally, Section 6 (Conclusions) summarizes the contributions of our study and reflects on its implications for super-resolution in CT imaging.

II. Method

Adopting the notation from the original SR4ZCT paper, we consider a reconstructed 3D volume from a phantom or a CT scan as $I(x, y, z) \in R^{X \times Y \times Z}$ where X, Y, Z are the numbers of pixels across the x, y and z axes. This volume can be visualized by slicing it across any of the three axes, thus viewing the plane of the other two dimensions. Slicing across the x,y,z axis yields sagittal, coronal and axial images in the YZ, XZ, and XY planes respectively at positions x', y', z'. Each of these images can be denoted as $s_{x'} = I(x', :, :), c_{y'} = I(:, y', :), a_{z'} = I(:, :, z')$. In real world CT images, the space that is covered by the voxel of the z axis r^z is usually larger than for x and y which are usually the same $r^x = r^y$. However, the distance d to the next voxel along the z axis might be smaller than the space covered by a single voxel which means that there is overlap along the z axis $o^z = r^z - d$. The task is then to leverage that overlap in order to increase the resolution across the z axis.

The core idea of SR4ZCT is to generate a new dataset of low resolution (LR) and high resolution (HR) images from the original CT volumes. The axial direction is used to get HR images whereas the LR ones are simulated as if they were affected by the through plane resolution reduction. This reduction is achieved by using an interpolation based down scaling function $F^x_\downarrow$ which is used on axial images a^x_i across the x axis with original pixel size $r^x \times r^y$ to generate images of size $r^z \times r^y$ with overlap of o^z between image rows. Then these down scaled images are brought back to their original dimensionality using an up scaling function $F^x_\uparrow$ which leads to a degraded version a_i' of the original axial image. Since the resolution is altered only across the z axis, these functions can also be reused to generate LR and HR data using the y axis by simply rotating the original image by 90 degrees before applying the functions.

By using this process a set of LR,HR pairs (a_i, a_i') can be generated where $a_i' = F_\uparrow(F_\downarrow(a_i))$, where a_i can be either the original or the rotated slice across the z axis. Those pairs can then be used to train a neural network f_θ which will learn to map a degraded image a_i' back to the original image a_i. Using a loss function L, the optimal weights of the network would be:

$$\theta^* = \arg\min_\theta \sum_i L(f_\theta(a_i'), a_i) \tag{1}$$

and the update rule using a single pair:

$$\theta \leftarrow \theta - \alpha \cdot \nabla_\theta L(f_\theta(a_i'), a_i) \tag{2}$$

After training is terminated the network has learned to map the LR images back to HR. Since the targeted area on the human body is usually targeted around a single area we can assume that the visual features captured by the CT images will be similar across different orientations. This means that the network f_θ can then be applied to the non artificially degraded sagittal and coronal slices, after up scaling using the $F_\uparrow$ function. The resulting images can be used as high resolution slices instead of the original sagittal and coronal images.

In this work we draw inspiration from the SR4ZTC algorithm and apply a similar version on our own 3D lung phantoms. We generate our phantoms from scratch, use ASTRA to generate sinograms and reconstructions of the phantoms to simulate a real CT scan, then process those reconstructions in the same manner as SR4ZTC to generate training data and apply the method using a UNET architecture.

To create a realistic low-resolution volume from our high-resolution phantoms, we simulate the entire CT data acquisition and reconstruction pipeline using the **ASTRA Toolbox**. This high-performance, GPU-accelerated library allows us to model geometric transformations and information loss that occur during the tomographic process. The simulation begins with a forward projection, where we model a parallel beam X-ray source rotating around the 3D phantom for 180 projection angles over a 360° arc. This process generates a high-resolution sinogram, which represents the raw data recorded by a virtual detector. Finally, this sinogram is reconstructed into a 3D volume using the robust Simultaneous Iterative Reconstruction Technique (SIRT). The result is a low-resolution volume that serves as a physically plausible input for our super-resolution task during inference.

For the **degradation function**, which simulates resolution loss on 2D slices for our self-supervised training set, we utilize the `scipy.ndimage.zoom` function. This function resizes an N-dimensional array using a specified order of spline interpolation. Our degradation process involves a two-step sequence. First, a high-resolution 2D slice is downsampled by a factor of four along a single axis (e.g., vertical) using the `zoom` function with the `order=1` argument, which specifies **bilinear interpolation**. This operation calculates the value for each new pixel in the smaller, low-resolution image by taking a weighted average of its four nearest neighbors in the original image, effectively modeling the signal integration of a physical detector element and causing a loss of high-frequency detail. Second, the small, low-resolution image is immediately upsampled back to its original dimensions, again using bilinear interpolation. This upsampling step cannot recover the lost details; instead, it smoothly interpolates between the existing low-resolution pixels, creating the characteristic blur that simulates the output of a low-resolution imaging system. This two-step process provides a computationally efficient method for generating the degraded inputs for our dataset.

The neural network architecture we selected for the image-to-image restoration task is the **U-Net**. The model `MSDRegressionModel` used in the original paper SR4CTZ was not able to be used due to hardware limitations such a CUDA version issue. UNet architecture is also well-suited for medical imaging tasks because of its symmetric encoder-decoder structure and characteristic skip connections. The encoder path progressively downsamples the input image through a series of convolutional and max-pooling layers, capturing hierarchical features at multiple spatial scales. At its deepest level, a bottleneck with a high feature capacity allows the network to learn complex, global contextual information. The decoder path symmetrically upsamples the feature maps, using transposed convolutions or interpolation, to reconstruct the image back to its original resolution. The critical innovation of the U-Net is the skip connections, which concatenate the feature maps from each level of the encoder directly to the corresponding level in the decoder. This allows the network to combine deep, semantic features from the decoder with shallow, fine-grained spatial details from the encoder, preventing the loss of high-frequency information and enabling the precise reconstruction of sharp edges and intricate textures, which is essential for super-resolution.

III. Experiments

In this section we describe how we generated the 3D phantoms and then the 2D data for our experiments. Then we explain the way we organized model training and evaluation.

A. 3D Lung Phantom Generation

A key challenge in training and evaluating super-resolution algorithms for medical imaging is the lack of high-resolution, isotropic, and co-registered ground-truth data. To address this, we developed a procedural 3D lung phantom generator designed to produce anatomically plausible, high-resolution volumes ($512 \times 512 \times 512$ voxels) for controlled experimentation. Implemented in Python and based on `PyVista` and `VTK`, our pipeline synthesizes phantoms through a sequence of geometry-driven steps that capture the variability and complexity of real lung structures.

The lungs are modeled as two asymmetric ellipsoids defined over a 3D NumPy grid, allowing for variation in size and orientation to reflect natural anatomical differences such as the cardiac notch. These volumetric shapes are converted to surface meshes using the marching cubes algorithm, producing `pyvista.PolyData` objects that serve as spatial constraints for the rest of the generation process.

A recursive algorithm is then used to construct the bronchial tree, beginning with a tracheal trunk that bifurcates into left and right bronchi. From there, branches are generated hierarchically, with each segment direction perturbed by a configurable degree of randomness to introduce realistic morphological variation. To ensure anatomical validity, the maximum length of each branch is computed based on its direction and proximity to the lung surface, preventing branches from extending outside the bounds of the lungs. We also implemented a fibre-growth mode which forces alignment along a single axis to produce patterns that are harder for the algorithm to detect since this information will be lost in the through plane resolution reduction.

Finally, the geometry is voxelized into a 3D array. Each branch segment is represented as a tapered cylindrical mesh, and the inclusion of each voxel is determined using `vtkImplicitPolyDataDistance`. We ensure fast estimation of the pixels contained in the mesh by only checking the cube generated by the minimum and maximum values of the x,y,z coordinates instead of the entire space. Intensity values are assigned based on structure type, and voxel coordinates are cached as `.npy` files to avoid redundant data generation. The result is a set of high-resolution, procedurally varied digital lung phantoms that serve as reproducible, verifiable ground truth for super-resolution experiments.

Using this generator, we created a diverse dataset of phantoms to rigorously test our model under various conditions. We systematically varied key parameters such as the growth pattern (standard vs. fibrous), and the primary axis of fibrous growth. This resulted in the following distinct categories of phantoms for our experiments:

- **Standard Phantoms (Non-Fibrous):** These phantoms feature a standard, organic branching pattern with a larger radius of 5 units.
- **Fibrous Phantoms:** To create a more challenging, anisotropic test case, we generated phantoms in "fibre" mode, where airway growth is heavily constrained along a single axis. These were also generated for each of the three primary axes (X, Y, and Z) with a starting radius of 5 units.
- **Fine-Structure Phantoms:** To evaluate the model's ability to resolve much finer details, we generated an additional set of standard, non-fibrous phantoms with a smaller branch starting radius of 2 units.

This combination of seven distinct phantom types allowed us to comprehensively assess our model's robustness and its performance on structures with varying complexity, scale, and anisotropy.

B. 2D Sinogram and Dataset Generation

To train and evaluate our super-resolution model on realistic CT data, we simulate a full tomographic pipeline using our generated 3D phantoms. We used the ASTRA Toolbox to perform high-fidelity forward projection and reconstruction. We used parallel-beam geometry that is defined for a full 360° rotation using 180 projection angles and a 512×512 detector. array ASTRA's GPU-accelerated `cuda3d` projector computes the sinograms which can be seen in Figure 2 along with the bronchi voxels. These sinograms are then downsampled along the detector's z-axis via local averaging to simulate the effect of thicker detector elements (4x resolution reduction). The resulting low-resolution sinograms are used to generate the reconstructions with ASTRA's implementation of the Simultaneous Iterative Reconstruction Technique (SIRT), yielding a final $512 \times 512 \times 128$ volume that closely mimics clinical through-plane degradation. This realistic degradation pipeline ensures that our learning framework operates on input data that simulate what a degraded CT scan of our phantom would look like but without any degradation along the z axis.

To train our network in a self-supervised fashion, we then need to generate training pairs in 2D from the high-resolution phantom volumes. Axial slices are extracted along the z-axis to serve as high-resolution targets (I_{HR}), while their low-resolution counterparts (I_{LR}) are generated using a custom degradation function. This function applies 1D down sampling by a factor of 4 followed by up sampling along a single axis using `scipy.ndimage` zoom function. This process simulates the degraded through-plane resolution seen real world CT scans. Each slice is normalized to [0, 255] and uninformative slices (e.g., empty lung tops) are excluded. For generalization, each HR slice is degraded twice: once along the vertical axis, and once along the horizontal. The horizontally-degraded images and their targets are rotated by 90°, ensuring the blur is always vertically aligned in the final dataset, simplifying the learning task.

Finally, we construct our training set by concatenating the vertically and rotated-horizontal degraded image pairs and randomly shuffling them. This strategy will expose the model to a broad set of directional blur patterns while enforcing a consistent canonical input form. By solving this simplified 2D problem, the network will learn to recover high-frequency details lost in anisotropic imaging. During inference, this skill will generalize to real sagittal and coronal slices of reconstructed low-resolution CT volumes. In order to limit the size of our generated dataset we used unsigned integers of 8 bits to store the voxel values scaling the data to the [0,255] range linearly instead of using floating point values.

C. Deep Learning Model Training and Evaluation

The final stage of our pipeline is the training and evaluation of a U-Net model designed for slice-wise super-resolution of CT volumes. We implemented a flexible 2D U-Net in PyTorch, structured with a dynamically defined encoder-decoder architecture. The encoder path, built from configurable feature channels, consists of repeated 3×3 convolution layers followed by max-pooling operations. A high-capacity bottleneck with doubled feature channels processes the most abstracted representation. The decoder path symmetrically upsamples using bilinear interpolation—chosen to avoid checkerboard artifacts—and concatenates encoder features via skip connections before refinement with further convolutions. A final

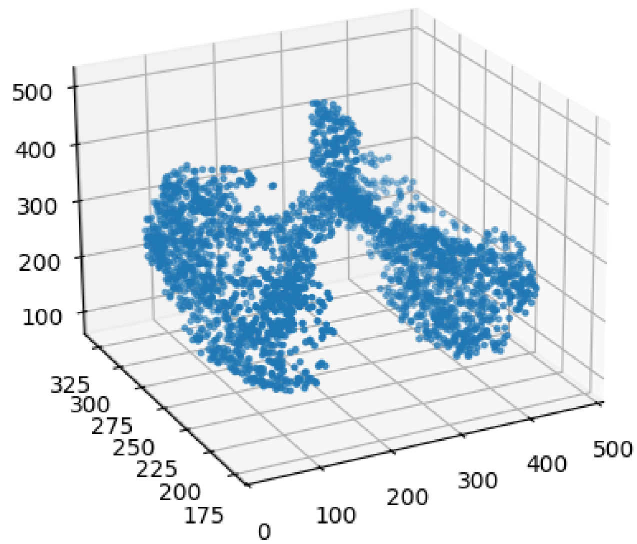

Fig. 1. Bronchi voxel representation

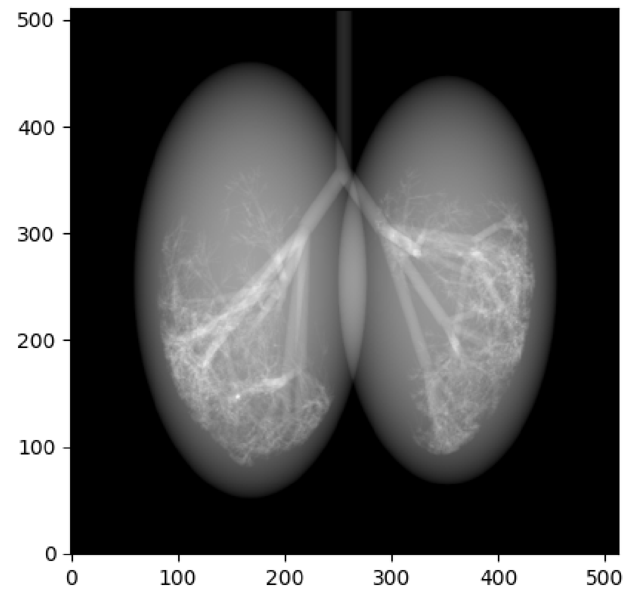

Fig. 2. Corresponding sinogram for single angle

1×1 convolution projects the multi-channel feature map into a single-channel grayscale output.

To rigorously test generalization across orientations, we adopt a principled data splitting strategy. The training set includes only coronal-plane slices from simulated phantoms, compelling the model to learn structural patterns from a single anatomical view. Input data is normalized, reshaped into tensors, and efficiently loaded with PyTorch's `DataLoader`, configured for shuffled mini-batches and parallel loading. The test set, in contrast, is built from axial and sagittal planes—orientations the model never sees during training. This separation ensures that performance evaluation reflects cross-plane generalization and not memorization. Deterministic loading (no shuffling) in the test loader guarantees consistency across evaluation epochs.

Model training proceeds via a supervised optimization loop using the Adam optimizer with a learning rate of 1×10^{-3}. The loss function is Mean Squared Error (MSE), computed between predicted and ground-truth slices. Each of the 50 training epochs involves standard steps: data loading, forward pass through the model, loss computation, backpropagation, and parameter updates. We facilitate reproducibility by using `argparse` to define phantom type, axis orientation, and normalization settings as command-line arguments. This modularity ensures adaptability across experiments and consistency in training protocols.

Evaluation is both qualitative and quantitative. At regular intervals, the model enters inference mode to produce visual outputs for a fixed subset of test images, displayed as triptychs of input, prediction, and ground truth. This visual log offers intuitive insights into performance progression. Quantitatively, the full test set is assessed using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM), capturing both pixel-level accuracy and perceptual quality. Scores are logged for baseline and predicted outputs. To preserve progress, model checkpoints are saved after each evaluation cycle, ensuring recoverability and enabling retrospective analysis of learned parameters.

To safeguard against interruptions and to preserve the state of the best-performing models, we implement a checkpointing strategy. After each evaluation cycle, the current state dictionary of the model's learned parameters (`model.state_dict()`) is serialized and saved to a file (`.pt`). This checkpointing ensures that we can always reload a model from a specific epoch for further analysis, fine-tuning, or final deployment without needing to retrain the entire model from scratch.

This training and evaluation method where data generation informs model training, and continuous, multi-faceted evaluation provides the feedback allows us to understand and validate the model's performance on this challenging self-supervised super-resolution task.

IV. Results and Analysis

Our comprehensive experimental setup, involving multiple phantom configurations and training variations, yielded a rich dataset of results. This section presents a detailed analysis of these findings, focusing on the model's overall performance, the impact of different phantom characteristics, the role of normalization, and the dynamics of the training process itself.

A. *Quantitative Performance Summary*

To establish a top-level understanding of our model's effectiveness, we first analyze the best performance achieved across all configurations. The results, summarized in Table I, compare the performance of our best model against the baseline low-resolution input. The "Best Overall Model" is identified as the run that achieved the highest PSNR improvement (from the `PSNR_Diff` column) at any epoch.

The data clearly demonstrates the success of our approach. The best-performing model, trained on a fibrous phantom with Y-axis orientation, achieved an exceptional PSNR of **41.57 dB**, representing a **2.44 dB** improvement over its already high-quality input of 39.13 dB. Across all experiments, the model consistently improved both PSNR and SSIM, with the best runs showing SSIM values reaching a near-perfect 0.99. This indicates that the U-Net not only improved pixel-wise fidelity but also masterfully restored the structural integrity of the images. While we do see a steady increase in the metrics when comparing with the input data, this increase does not reach the levels of the original paper. We estimate that this happens due to the sparse nature of our phantom based training data as well as the smaller complexity of our neural networks architecture, compared to the SR4ZTC model selection.

B. *Impact of Phantom Characteristics on Performance*

Our procedural phantom generator allowed us to create phantoms with varying structural properties. To systematically analyze the model's performance on these variations, we compiled the best-achieved PSNR improvement for each phantom type, as shown in Table I.

Several key trends emerge from this data:

- **Fibre vs. Non-Fibre Phantoms:** A consistent trend is evident in Table I. The models trained on fibrous phantoms consistently achieved the highest PSNR improvements**. The top-performing run on any fibrous phantom (+2.44 dB) significantly outperformed the top-performing run on any non-fibrous phantom (+1.61 dB). This suggests that the highly structured, repetitive, and anisotropic patterns in the fibre phantoms may provide a clearer, more easily learnable degradation signal for the network. The model can effectively learn the specific blur kernel for these oriented features and reverse it, leading to very high performance.
- **Influence of Fibre Axis (X, Y, Z):** Among the fibrous phantoms, the orientation of the fibres had a significant impact. The Y-axis oriented phantom proved to be the "easiest" for the model to learn, yielding the highest gains across both normalization schemes. The X-axis phantoms were more challenging, and the Z-axis phantoms were the most difficult of the fibrous set. This is an interesting finding, likely related to how the degradation (always applied vertically and horizontally in the 2D slice) interacts with the inherent orientation of the phantom's structures. When the fibres align favorably with the degradation axes, the learning task appears to be simplified.

C. *Analysis of Normalization and Training Dynamics*

We investigated two training strategies: one where the input data was normalized to a [0, 1] range ("Norm: Yes") and one where it was left in the [0, 255] range ("Norm: No"). We also analyzed the difference between performance at the final epoch versus the best-performing epoch.

- **Effect of Normalization:** The results, visible in Table I, do not show a universally superior strategy. In most cases, training without normalization yielded a slightly better peak result. The most notable observation is from the training loss: models trained with normalization ("Norm: Yes") converged to a very low, stable training loss (e.g., from the `Train_Loss` column, a value of 0.0001), while models without normalization had a much higher and more variable training loss (e.g., 5.3792). This indicates that while normalization stabilizes the training process and makes the loss values more interpretable, it does not necessarily guarantee a better final validation performance in terms of PSNR or SSIM for this specific architecture and task.
- **Overfitting and the "Best" Epoch:** In almost every single experiment, the peak performance (highest `PSNR_Diff`) occurred relatively early in training, often between epochs 8 and 17. By epoch 50, the performance had decayed from its peak. For example, the best model achieved a +2.44 dB gain at epoch 9, but this decayed to +1.53 dB by epoch 50. This is a classic sign of *overfitting*. The model learns the general restoration task well in the early epochs, but as training continues, it begins to memorize fine-grained noise and artifacts specific to the training set (coronal slices), which harms its ability to generalize to the unseen test set (axial and sagittal slices). This is clearly illustrated by the learning curve for one of the experiments, shown in Figure 3. This finding strongly suggests that implementing an **early stopping** mechanism—where training is halted when performance on a validation set no longer improves—would be a crucial step for optimizing the final model.
- **The Case of Negative Performance:** One experiment stands out: the `Radius 2 | Fibre: False | Axis: X | Norm: Yes` run. At its best epoch (11), it achieved a respectable +1.02 dB improvement. However, by epoch 50, its performance had catastrophically collapsed, yielding a `PSNR_Diff` of **-0.43 dB**. The model at this stage was producing an output that was quantifiably worse than the low-resolution input. This severe overfitting highlights the instability of prolonged training without regularization or early stopping, especially on potentially more complex or less distinct structures (smaller radius, non-fibrous).

D. *Qualitative Visual Analysis*

Beyond the quantitative metrics, a visual inspection of the output images provides further insight. Figure 4 shows a side-by-side comparison for a representative test slice from our best-performing model. The input image (a) is visibly blurry, particularly around the fine bronchial walls. The output from our U-Net (b) is substantially sharper, with clearer edge definition and better-resolved small structures, closely resembling the ground truth (c). This visual evidence confirms that the high PSNR/SSIM scores correspond to a genuine and meaningful improvement in perceivable image quality. We also include more images from our models in the appendix.

TABLE I
BEST ACHIEVED PERFORMANCE ACROSS ALL PHANTOM VARIETIES

			Normalization: No			Normalization: Yes		
Radius	**Phantom Type**	**Axis**	**PSNR Diff ↑**	**Out PSNR ↑**	**Out SSIM ↑**	**PSNR Diff ↑**	**Out PSNR ↑**	**Out SSIM ↑**
5	`Fibre: True`	X	1.76	38.15	0.98	1.53	37.92	0.97
5	`Fibre: True`	Y	**2.44**	41.57	0.99	2.33	41.46	0.99
5	`Fibre: True`	Z	1.36	40.10	0.98	1.31	40.05	0.98
5	`Fibre: False`	-	1.61	36.94	0.97	1.48	36.81	0.97
5	`Fibre: False`	-	1.42	36.17	0.97	1.41	36.16	0.96
5	`Fibre: False`	-	1.55	36.93	0.97	1.46	36.84	0.96
2	`Fibre: False`	-	1.14	37.16	0.97	1.02	37.04	0.97

Metrics correspond to the epoch with the highest PSNR improvement for each configuration.

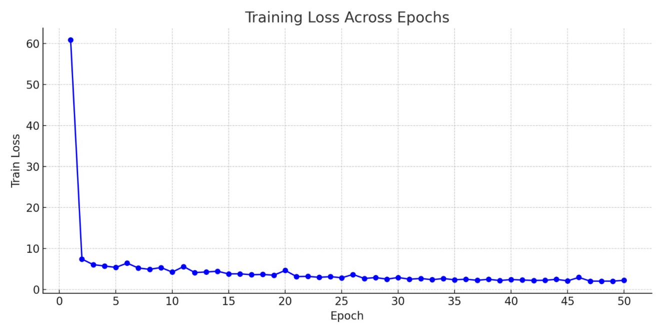

Fig. 3. Learning curve for the `Radius 5, Fibre: True, Axis: Y, Norm: No` experiment. The PSNR improvement on the test set (blue line) peaks at epoch 9 and then gradually decays, demonstrating overfitting as the training loss (orange line, secondary axis) continues to decrease.

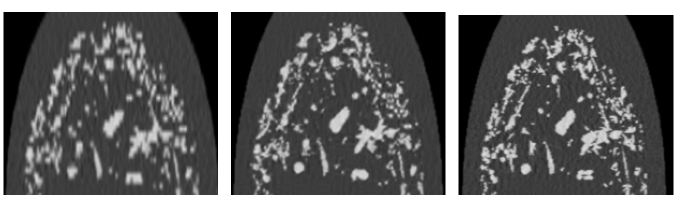

Fig. 4. Visual comparison between low-resolution input (left), model output (center), and high-resolution ground truth (right).

V. CONCLUSION AND FUTURE WORK

In this study, we presented a fully self-supervised pipeline for super-resolving through-plane CT slices using a dynamically configurable 2D U-Net architecture. By leveraging off-axis views of synthetically generated phantoms, we demonstrated that meaningful spatial priors can be learned from the other axis information. Our training protocol, and our evaluation strategy across orthogonal planes confirmed the model's ability to generalize beyond the training view. The results, both qualitative and quantitative, indicate that self-supervised learning from within-volume geometry can yield high-fidelity reconstructions suitable for resolution enhancement tasks in Computational Tomography.

Looking forward, several avenues remain open for exploration. First, we intend to extend the current framework to real clinical CT data, perhaps using the dataset from the original paper. Additionally, we aim to overcome CUDA compatibility issues in order to utilize the original paper's model and compare the performance when using a more sophisticated neural network architecture.

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VI. APPENDIX

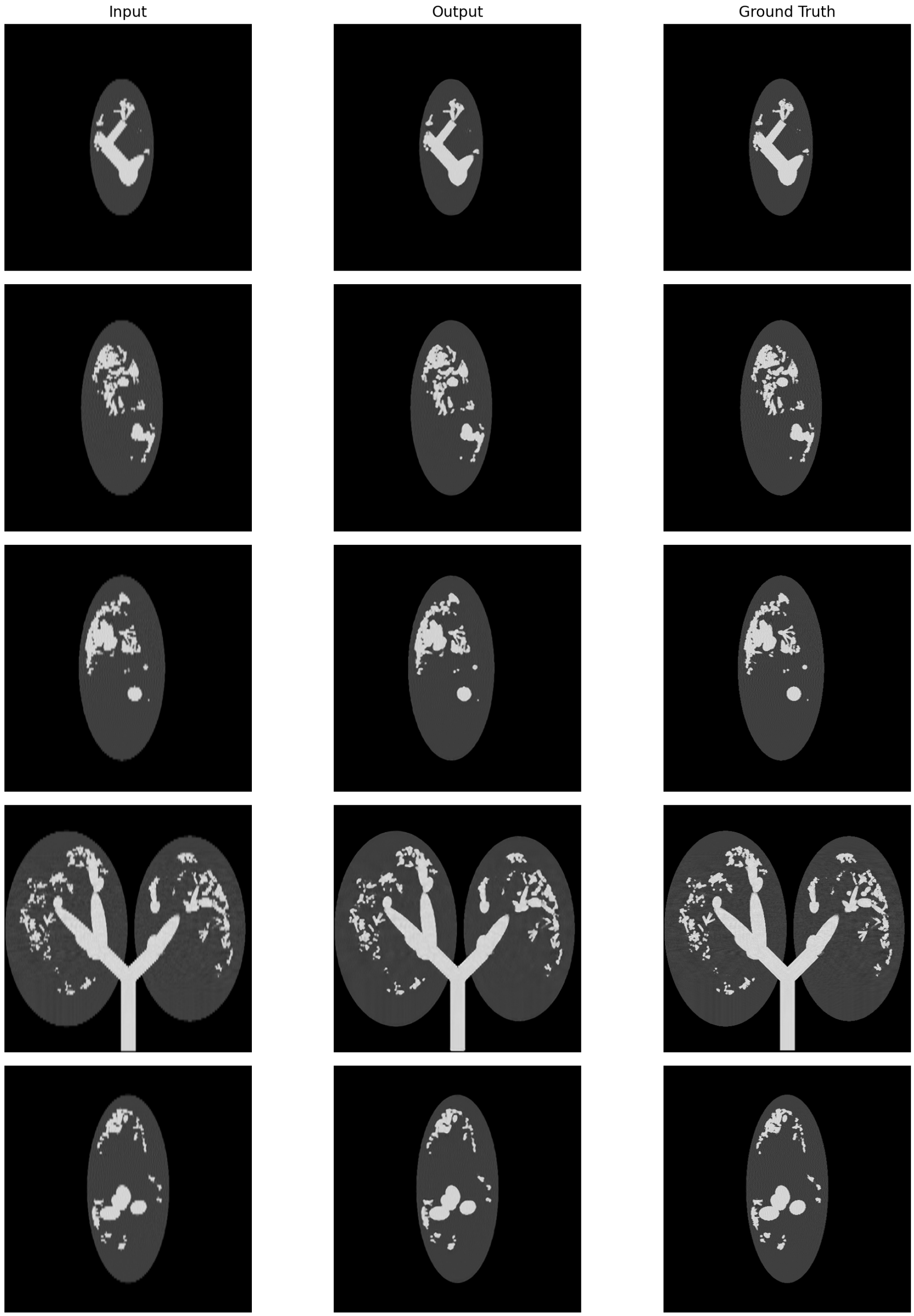

Fig. 5. Visual comparison on a test slice from the best-performing model run. (a) Low-resolution input. (b) Super-resolved output from our U-Net. (c) High-resolution ground truth. The model successfully restores sharp edges and fine details of the bronchial tree.